

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS, NOVEMBER 2025

Quantum Information Theory [CSE_40B3]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the measurements associated with mutual information (4)
A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
B) Compute $I(X;Y)$ if $H(X)=2$, $H(Y)=1.5$, and $H(X,Y)=2.8$. (4)
Course Outcome: "CO1" / Cognitive Level: "Application"
C) Examine why is mutual information always non-negative (2)
Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Explain density matrix (4)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) Construct the density matrix of pure state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ (4)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Examine How do density matrices describe mixed states (2)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Explain Teleportation (4)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Identify resources needed for teleportation (4)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) Examine the reasons for classical communication still being necessary (2)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Explain BB84 protocol (4)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

B) Identify the two bases which are used to encode bits in BB84 (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

C) List reasons Why does basis mismatch lead to discarding bits (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5) Explain Syndrome (4)

A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Identify Information being provided by syndrome (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

C) List reasons why must syndrome measurements avoid collapsing logical state (2)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	20(40%)
4	Analysis	10	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)